

The Schatten interpolation conjecture for Euclidean sections of ℓ_p -balls

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Status

This is a claimed full solution, subject to human review, of Conjecture 3.3(b) from Szarek–Wolff [1]. Since the source states that Conjecture 3.3(a), (b), and (c) are formally equivalent, this also gives the operator-ideal ingredient requested there for the sharp $1 < p < 2$ radius lower bound.

Source Problem

The source paper is S. J. Szarek and P. Wolff, *Radii of Euclidean sections of ℓ_p -balls*, arXiv:2410.15118 [1]. In Section 3, after proving the $p = 1$ endpoint through Pietsch’s lemma $\pi_2(i_{12}) = 1$, the authors ask for the corresponding sharp $1 < p < 2$ statement.

meier and Pajor [20], one deduces that

$$(3.2) \quad \lambda' \geq c d^{1/p-1/2},$$

and there are several other lines of argument that yield a similar conclusion. We conjecture that inequality (3.2) holds with $c = 1$. However, a proof appealing to [26] (i.e. via comparison of volumes of B_p^d - and B_2^d -balls) is clearly suboptimal; this is true already in the case $p = 1$, see Remark 2.4(a). A natural way to generalize Theorem 2.1(ii) to $1 < p \leq 2$ with a sharp estimate on λ' would be to find a counterpart for Lemma 2.3. Below we propose a few statements as problems/conjectures; they correspond to the different proofs of Lemma 2.3 we presented in Section 2:

Conjecture 3.3. (a) $\pi_{r,2}(i_{p2}) \leq 1$ for $p \in (1, 2)$, where $i_{p2} : \ell_p \rightarrow \ell_2$ is the formal identity map, $\pi_{r,2}(\cdot)$ is the $(r, 2)$ -absolutely summing norm (see Chapter 2 of [34]), and $r = 2p/(2 - p)$.

(b) $\|B : \ell_2 \rightarrow \ell_p\| \geq \|B\|_{S_r}$ with $r = 2p/(2 - p)$, where $\|A\|_{S_r} = (\text{Tr}(A^*A)^{r/2})^{1/r}$ is the r -Schatten norm.

(c) If A is positive semi-definite, then $\|A : \ell_{p'} \rightarrow \ell_p\| \geq \|A\|_{S_r}$ with $r = p/(2 - p)$ (and, again, $p' = p/(p - 1)$).

Confirming any of these conjectures would imply a generalization of Theorem 2.1(ii) to $p \in (1, 2)$ with the (optimal) assertion $\lambda' \geq d^{1/p-1/2}$. (Actually, like in the case of the different reformulations of Lemma 2.3, the statements (a)–(c) are formally equivalent.) Note that the dimension doesn't explicitly enter any of the statements (a)–(c) above, they are just assertions about various operator ideal norms and/or (real or complex) matrices of arbitrary size. It is also conceivable that there is a simple direct proof of the inequality (3.2) with $c = 1$, either via change of density (see Section 1.2 in [19]) or by

The formulation used here is Conjecture 3.3(b): for $1 < p < 2$ and

$$r = \frac{2p}{2 - p},$$

every finite real or complex matrix B , viewed as an operator $\ell_2 \rightarrow \ell_p$, should satisfy

$$\|B : \ell_2 \rightarrow \ell_p\| \geq \|B\|_{S_r}.$$

Proof Intuition

The conjecture is an interpolation statement in disguise. Dualizing changes $B : \ell_2 \rightarrow \ell_p$ into $C = B^* : \ell_q \rightarrow \ell_2$, where $q = p'$. The desired Schatten exponent is exactly the exponent obtained by interpolating between the two endpoint inclusions

$$\mathcal{L}(\ell_2, \ell_2) \hookrightarrow S_\infty \quad \text{and} \quad \mathcal{L}(\ell_\infty, \ell_2) \hookrightarrow S_2.$$

The first endpoint is just the equality between Hilbert operator norm and largest singular value. The second endpoint is Pietsch's $p = 1$ mechanism in one line: averaging $C\varepsilon$ over signs recovers the Hilbert–Schmidt norm. Interpolation then supplies the exact constant 1 for all $2 < q < \infty$.

Theorem

Theorem. Let $1 < p < 2$, and set $r = 2p/(2 - p)$. For every finite matrix B over $\mathbb{R}$ or $\mathbb{C}$,

$$\|B\|_{S_r} \leq \|B : \ell_2 \rightarrow \ell_p\|.$$

Proof. It is enough to prove the dual form. Put $q = p'$, so $2 < q < \infty$, and notice that

$$\frac{1}{r} = \frac{1}{p} - \frac{1}{2} = \frac{1}{2} - \frac{1}{q}.$$

Since $\|B : \ell_2 \rightarrow \ell_p\| = \|B^* : \ell_q \rightarrow \ell_2\|$ and $\|B\|_{S_r} = \|B^*\|_{S_r}$, it suffices to show that every finite matrix $C : \ell_q^n \rightarrow \ell_2^m$ satisfies

$$\|C\|_{S_r} \leq \|C : \ell_q^n \rightarrow \ell_2^m\|.$$

We prove this by interpolation. At the Hilbert endpoint,

$$\|C\|_{S_\infty} = \|C : \ell_2^n \rightarrow \ell_2^m\|.$$

At the other endpoint, let $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)$ be a vector of independent Rademacher signs. Since $\|\varepsilon\|_\infty = 1$,

$$\|C : \ell_\infty^n \rightarrow \ell_2^m\|^2 \geq \mathbb{E} \|C\varepsilon\|_2^2 = \sum_{j=1}^n \|Ce_j\|_2^2 = \|C\|_{S_2}^2.$$

Thus the identity map on $m \times n$ matrices is contractive as a map

$$\mathcal{L}(\ell_2^n, \ell_2^m) \longrightarrow S_\infty^{m,n}$$

and also as a map

$$\mathcal{L}(\ell_\infty^n, \ell_2^m) \longrightarrow S_2^{m,n}.$$

Let $0 < \theta < 1$ be chosen by

$$\frac{1}{q} = \frac{1 - \theta}{2}, \quad \text{equivalently} \quad \frac{1}{r} = \frac{\theta}{2}.$$

By the standard exact interpolation identities for finite-dimensional ℓ_p spaces, Hilbert-valued operator spaces, and Schatten classes [2, 3],

$$[\mathcal{L}(\ell_2^n, \ell_2^m), \mathcal{L}(\ell_\infty^n, \ell_2^m)]_\theta = \mathcal{L}(\ell_q^n, \ell_2^m)$$

and

$$[S_\infty^{m,n}, S_2^{m,n}]_\theta = S_r^{m,n},$$

isometrically. Interpolating the identity map therefore gives

$$\|C\|_{S_r} \leq \|C : \ell_q^n \rightarrow \ell_2^m\|.$$

This proves the dual form and hence the desired inequality for B . □

Consequences

Corollary. For $1 < p < 2$ and $r = 2p/(2 - p)$, the formal identity $i_{p2} : \ell_p \rightarrow \ell_2$ satisfies

$$\pi_{r,2}(i_{p2}) \leq 1.$$

In fact equality holds.

Proof. Using the standard finite-rank characterization of the $(r, 2)$ -summing norm, the theorem says exactly that

$$\|i_{p2}T\|_{S_r} \leq \|T : \ell_2 \rightarrow \ell_p\|$$

for every finite-rank T . Taking the supremum over $\|T\| \leq 1$ gives $\pi_{r,2}(i_{p2}) \leq 1$. Rank-one operators give equality. $\square$

Verification Notes

The proof uses only endpoint inequalities and standard interpolation. The Rademacher endpoint is valid over both real and complex scalars, since the signs are real and $\mathbb{E} \varepsilon_i \varepsilon_j = \delta_{ij}$. The interpolation identities used above are finite-dimensional and hold for real spaces by the usual real-form version of the complex interpolation theorem, or by applying the same argument to complexifications and restricting back to the real part.

Before promotion, the run indexes were searched for arXiv:2410.15118, the paper title, $\pi_{r,2}(i_{p2})$, and the $\ell_2 \rightarrow \ell_p$ Schatten formulation. No exact prior packet or attempt was found. Exact web searches for the same phrases found no later direct answer. A numerical finite-matrix probe is included only as sanity checking; it is not used in the proof.

Limitations

The packet proves the operator-ideal formulation Conjecture 3.3(b) and the $(r, 2)$ -summing formulation Conjecture 3.3(a). The source paper states that the three formulations in Conjecture 3.3 are formally equivalent and that confirming any one of them implies the sharp $c = 1$ lower bound $\lambda' \geq d^{1/p-1/2}$. This packet relies on that source-stated equivalence for the geometric restatement, and does not rederive the source paper's reduction from the operator inequality to section radii.

References

References

- [1] S. J. Szarek and P. Wolff, *Radii of Euclidean sections of ℓ_p -balls*, arXiv:2410.15118.
- [2] J. Bergh and J. Lofstrom, *Interpolation Spaces. An Introduction*, Springer-Verlag, 1976.
- [3] B. Simon, *Trace Ideals and Their Applications*, 2nd ed., American Mathematical Society, 2005.